# TITLE OF THE DISSERTATION IN ALL CAPITAL LETTERS

by

**Candidate Full Name**

DISSERTATION

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Approved:

| | |
|---|---|
| Dissertation Chair | Date |
| Committee Member | Date |

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by

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# DEDICATION

Dedicated to my mentors, colleagues, family, and friends who supported this journey.

# ACKNOWLEDGEMENTS

I express my deepest gratitude to my advisor, dissertation committee members, and laboratory colleagues.

# CONTRIBUTIONS

Descriptions of student and co-author contributions for adapted manuscript chapters.

# EPIGRAPH

*"Equipped with his five senses, man explores the universe around him and calls the adventure Science."*

— Edwin Powell Hubble (1948)

# ABSTRACT

**TITLE OF THE DISSERTATION IN ALL CAPITAL LETTERS**

**AUTHOR FULL NAME IN ALL CAPITAL LETTERS**

A concise synthesis of the research problem, methodology, findings, and conclusions. The abstract must be double-spaced and contain no abbreviations in the title.

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# LIST OF ABBREVIATIONS

**DNA** Deoxyribonucleic Acid

**RNA** Ribonucleic Acid

**UCSF** University of California, San Francisco

# CHAPTER 1

# INTRODUCTION

## 1.1 Background and Motivation

Biological research in the modern era relies increasingly on multi-omics profiling and computational data modeling. In this chapter, we outline the foundational principles and state-of-the-art literature motivating this dissertation [1].

## 1.2 Heading Hierarchy Example

As required by the institutional formatting rubric (Sec III.J), headings must strictly adhere to the four-level hierarchy:

1. **Primary Headings** (`\chapter{}`): ALL CAPITAL LETTERS, bold, centered, double spaced above major sections.
2. **Secondary Headings** (`\section{}`): Sentence Case, bold, left-aligned.
3. **Tertiary Headings** (`\subsection{}`): Sentence Case, italicized, unbolded, left-aligned.
4. **Quaternary Headings** (`\subsubsection{}`): Sentence Case, italicized, in-line with paragraph, followed by a period.

### *1.2.1 Tertiary Subsection Heading Example*

Tertiary headings are italicized and unbolded in sentence case. Paragraph text begins double-spaced underneath.

*Quaternary Heading Example.* Quaternary headings run directly in line with paragraph text, followed immediately by a period.

## 1.3 Table Formatting Example

Per Rubric Sec III.A, table titles/captions must be placed **above** the table, single-spaced, with bolded table references.

**Table 1.1:** Demographic summary of study cohorts evaluated across chapters.

| Cohort Name | Ancestry | Sample Size (N) | Platform |
|---|---|---|---|
| Discovery Cohort | Multi-Ancestry | 621 | EPIC BeadChip |
| Replication Cohort A | African American | 1,394 | 450K BeadChip |
| Replication Cohort B | European | 729 | 450K BeadChip |

## 1.4 Figure Formatting Example

Per Rubric Sec III.B, figure titles and legends must be placed **below** the figure and single-spaced.



**Figure 1.1: Schematic overview of the analytical study workflow.** High-throughput DNA methylation and genomic variant data are preprocessed, followed by ancestry-stratified evaluation of epigenetic clock predictions and methylation quantitative trait loci (meQTL) mapping.

## 1.5 Block Quotation Formatting Example

Per Rubric Sec I.F and III.C, prose quotations over 4 lines must be single-spaced and indented uniformly:

> Biological aging is a complex, multifaceted process characterized by the gradual deterioration of physiological function and increased susceptibility to disease. Epigenetic modifications, particularly DNA methylation at cytosine-phosphate-guanine dinucleotides, provide a dynamic and quantifiable record of cellular chronological and physiological history.

Double-spaced prose continues here immediately following the quotation block.

# REFERENCES

[1] Steve Horvath. “DNA methylation age of human tissues and cell types”. In: *Genome Biology* 14.10 (2013), p. 3156. DOI: 10.1186/gb-2013-14-10-r115.

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